

SAJIVAN KUNARETHINAM

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EDUCATION

Sri Lanka Institute of Information Technology (SLIIT)

Bachelor of Science Honours in Information Technology

Specialized in Data Science (CGPA: 3.72 / 4.0)

Expected: June 2027

- **Relevant Coursework:** Supervised Learning, Unsupervised Learning, Machine Learning & Optimization Methods, Fundamentals of Data Mining, Deep Learning, Artificial Intelligence, Information Retrieval & Web Analysis, Statistical & Predictive Modelling, Database Systems, Programming with Python, Massive Data Processing and Cloud Computing
- **Dean's List Award** – 1st Year 2nd Semester, 2nd Year 1st Semester, 2nd Year 2nd Semester

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript, Java, C++,
- **Machine Learning:** scikit-learn, TensorFlow, PyTorch, XGBoost, PySpark MLlib, Random forest
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Next.js, Express.js
- **Frameworks & Tools:** FastAPI, Streamlit, MongoDB, Supabase, Git
- **Cloud & Big Data:** Azure, AWS (basics), Spark, Hadoop
- **Core Concepts:** Regression & Classification, Neural Networks, Deep Learning (CNNs), Feature Engineering, Model Evaluation (R^2 , AUC, F1), Explainable AI (SHAP), MLOps fundamentals

PROJECTS

Medical Insurance Premium Prediction – TensorFlow/Keras · scikit-learn · Streamlit | [GitHub](#) | [Streamlit app](#)

- Built an ANN regressor (128,64,1, MSE loss) with proper scaling for X and y to predict insurance charges; achieved $\sim 70\%$ R^2 on test
- Shipped an interactive Streamlit app with packaged artifacts (insurance_model.keras, scaler_x.pkl) for instant predictions from user inputs.
- Implemented a clean ML pipeline: one-hot encoding, StandardScaler/MinMaxScaler, reproducible requirements, and README docs for quick run.

Vehicle CO₂ Emission Predictor – scikit-learn · XGBoost · MLR · Streamlit | [GitHub](#) | [Streamlit app](#)

- Achieved $R^2 \approx 0.86$ on hold-out vehicle emission data by engineering interaction features and ensembling (Random Forest + XGBoost).
- Delivered an interactive Streamlit app for real-time CO₂ predictions from vehicle specs (engine size, cylinders, fuel consumption) and packaged model + UI for public use.
- Built a robust data pipeline: cleaned >100K rows, handled multicollinearity, encoded categorical variables, and documented workflows for reproducibility.

Employee Attrition Prediction App – scikit-learn · XGBoost · SHAP · Streamlit | [GitHub](#) | [Streamlit app](#)

- Improved employee churn detection recall from 0.61 $\rightarrow$ 0.74 and achieved ROC-AUC = 0.80 / Brier Score = 0.1057 by calibrating Logistic Regression and tuning decision thresholds to reduce missed leavers
- Deployed an interactive Streamlit app with SHAP-based explainability and engineered features like Years Per Company & Income Per Year, enabling HR teams to visualize attrition risk and understand key factors (Overtime, Income, Travel Frequency)

Password Safety AI System – FastAPI · Next.js · MongoDB · Agents | [GitHub](#)

- Boosted strong-password adoption to $\approx 78\%$ among test users by shipping a multilingual passphrase generator (Sinhala/Tamil/English) and integrating a breach-check via the HaveIBeenPwned API, improving overall account security posture
- Designed and deployed a multi-agent system comprising Generator, Watchdog, Vault, and Orchestrator that automates password lifecycle management, domain-aware de-duplication, and secure autofill readiness – enabling scalable password safety operations.

Fraud Detection Prediction App – scikit-learn · Streamlit · Pandas · Joblib | [GitHub](#) | [Streamlit app](#)

- Achieved **94% accuracy** and **0.94 recall** by training a Logistic Regression model on a **1.9M+ transaction dataset** with a severe 773:1 class imbalance, leveraging `class_weight='balanced'` for robust fraud detection.
- Built and deployed an interactive **Streamlit web app** enabling users to input transaction details and obtain real-time fraud predictions with probability confidence, improving interpretability and accessibility for end-users.

CERTIFICATIONS & ACHIEVEMENTS

Programming in Python (meta) – [CERTIFICATE](#)

Supervised Machine Learning (Stanford) – [CERTIFICATE](#)

Google AI Essential (Google) – [CERTIFICATE](#)

Machine Learning PySpark (Coursera) – [CERTIFICATE](#)

Linear Regression with Python (Coursera) – [CERTIFICATE](#)

CNNs with TensorFlow (Coursera) – [CERTIFICATE](#)

100 % Academic Scholarship (2025–2026) – SLIIT Merit Award